Simulation methods in Medical Engineering
Spring 2025
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CM2024 Mathematical and numerical background

Lecture notes for lecture 3

IVPs continued

Boundary value prolems

Stiff and non-stiff IVPs

Overview of Lab 2

Instructions for the examination of Lab 1 and Lab 2
Deadline for submission: March 3rd 2025

First of all, for all your submitted MATLAB-programs: Put a comment (%text…….) in the beginning of your programs with the NAMES of the two group members!

On Thursday Jan 23rd, 13-17, there is Lab-time in Gullbrandssalen and you can ask questions.

For other times:

1) If you need help with your MATLAB program(s) you e-mail the corresponding m-file to edsberg@kth.se together with your questions.
I will then inspect your program(s) and send it back to you with modifications and/or comments.

2) If you want to finish your completed Labs with **Pass** you send via Canvas the working MATLAB-programs and the graphs of the results as pdf-files. If you aspire to **Pass with Excellence** you also submit the report as a pdf-file. The requirements for Pass and Pass with Excellence are described in the notes of Lecture 1.

3) It is OK to submit your ready Labs any time between Jan 23rd - March 3rd.

Some final remarks about till IVP.
As Lab 1 shows different methods are tried for the numerical solution of MP2, the harmonic oscillator.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0,\ x(0)=1,\ \frac{dx}{dt}(0)=0 \quad (*)$$

Some of the methods work satisfactory, some not, which is illustrated by running lab1demo.m. This m-file is to be expanded by you with some more methods.

The IVP (*) can be written in matrix form

$$\frac{d\bar{u}}{dt} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}\bar{u},\ \bar{u}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

1) With this form we can also apply ode45. This is done in the m-files harmosc.m fharmosc.m. We see that the energy $H(t) = (\dot{x}(t))^2 + \omega^2 (x(t))^2$ is not satisfactorily preserved.

2) Finally (*) can be solved by using central difference, MATLAB-program harmoscentral.m

$$\frac{x_{k+1} - 2x_k + x_{k-1}}{h^2} + \omega^2 x_k = 0,\ x_0 = 1,$$

$$x_1 \approx x(h) = x(0) + h\dot{x}(0) + \frac{h^2}{2}\ddot{x}(0) + O(h^3)$$

$$x_1 = 1 + \frac{h^2}{2}(-\omega^2) = 1 - \frac{h^2\omega^2}{2}$$

$$x_{k+1} = (2 - h^2\omega^2)x_k - x_{k-1},\ k = 1, 2, \ldots, N-1$$

h must be in the interval $0 < h \leq 1$ for stability

Boundary Value Problems, BVPs.

In a BVP you state the boundary conditions instead of the initial conditions.
In many cases you have a second order ODE given, together with two boundary conditions.

MP3 (see Lecture notes Lecture 1)

$$-\frac{d^2u}{dx^2} + a\frac{du}{dx} = f(x),\ u(0)=\alpha,\ u(1)=\beta \qquad (*)$$

Formulate as an IVP? We don't know $u'(0)$!

Formulate as a system

$v_1 = u$, $v_2 = \frac{du}{dx}$, Let $\alpha=1, \beta=0, a=10, f=1$

$$\frac{dv_1}{dx} = v_2 \qquad v_1(0) = 1$$

$$\frac{dv_2}{dx} = 10v_2 - 1 \qquad v_2(0) = k, \text{ the slope at } x=0$$

$$\frac{d\bar{v}}{dx} = \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 10 \end{pmatrix}}_{A}\bar{v} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \bar{v}(0) = \begin{pmatrix} 1 \\ k \end{pmatrix} \qquad (**)$$

Try different values of k: shoot.m, fshoot.m
This is called a shooting method.
Hard to find k, $u(1)$ is very sensitive in k.

The eigenvalues of A are $\lambda_1 = 0$, $\lambda_2 = 10$
$\Rightarrow$ there is one part of the solution growing as e^{10x} $\Rightarrow$ The IVP (**) is unstable

Try another method (LE 4.2.1) $\Rightarrow$

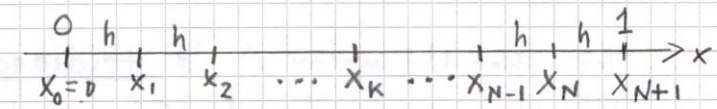
Approximate the derivatives in MP3, by difference quotients (se Lecture 1)

$$\frac{du}{dx}(x_i) = \frac{u(x_{i+1}) - u(x_{i-1})}{2h} + O(h^2)$$

$$\frac{d^2u}{dx^2}(x_i) = \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2} + O(h^2)$$

These central differences work on BVPs but usually not on IVPs, but is successful only if h small
Skip the $O(h^2)$-terms and insert into the ODE:

at $x = x_k$: $$-\frac{u_{k+1} - 2u_k + u_{k-1}}{h^2} + a\frac{u_{k+1} - u_{k-1}}{2h} = f(x_k) \quad (*)$$



x_0 and x_{N+1} are boundary points
$x_1, x_2, \ldots, x_N$ are inner points.
At $x_0 = 0$, u_0 is known $= \alpha$
At $x_{N+1} = 1$, u_{N+1} is known $= \beta$
At $x_1, x_2, \ldots, x_K, \ldots, x_N$, u_k, $k = 1, 2, \ldots, N$ is unknown.

$(*)$ is valid at all inner points, $k = 1, 2, \ldots, N$
Hence we have N unknowns and N eqns

$k = 1$: $$-\frac{u_2 - 2u_1 + \cancel{u_0}^{\alpha}}{h^2} + a\frac{u_2 - \cancel{u_0}^{\alpha}}{2h} = f(x_1)$$

$k = N$: $$-\frac{\cancel{u_{N+1}}^{\beta} - 2u_N + u_{N-1}}{h^2} + a\frac{\cancel{u_{N+1}}^{\beta} - u_{N-1}}{2h} = f(x_N)$$

Keep all unknowns in the left hand side and all known terms in the right hand side
Multiply all eqns with h^2:

$$\underbrace{\begin{pmatrix} 2 & -1+\frac{ha}{2} & 0 & \cdots & 0 \\ -1-\frac{ha}{2} & 2 & -1+\frac{ha}{2} & & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & & -1-\frac{ha}{2} & & 2 \end{pmatrix}}_{A,\ N\times N} \underbrace{\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{N-1} \\ u_N \end{pmatrix}}_{\text{unknowns } \bar{u}} = \underbrace{\begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix}}_{f(x)} + \underbrace{\begin{pmatrix} \alpha + \frac{h\alpha\alpha}{2} \\ 0 \\ \vdots \\ 0 \\ \beta - \frac{h\alpha\beta}{2} \end{pmatrix}}_{BVs}$$

This is a linear system of algebraic eqns

$$A\bar{u} = \bar{b}$$

which is solved by Gaussian elimination, GE in MATLAB

$$u = A \backslash b$$

Note that the matrix A is <u>tridiagonal</u>. The general algoritm for GE can then be designed much more efficiently.

The solution of MP3, is performed by the MATLAB-program <u>bvp.m</u>

Demonstration of the numerical solution of a partial differential eqn, PDE, although PDEs are not included in this part 1 of the course. The concept of stiffness, however, is within the course (Definition in the middle of LE pg 53):

* $\mathrm{Re}(\lambda_i) \leq 0$
* $\mathrm{Re}(\lambda_i)$ are of very different size

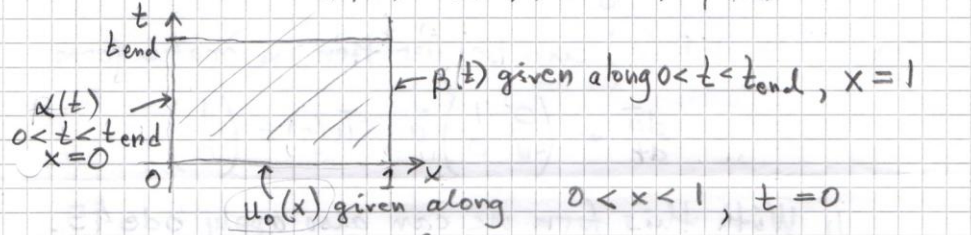
As a last example of numerical solution of IVPs, let's look at the heat equation

PDE: $\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $0 < t < t_{end}$

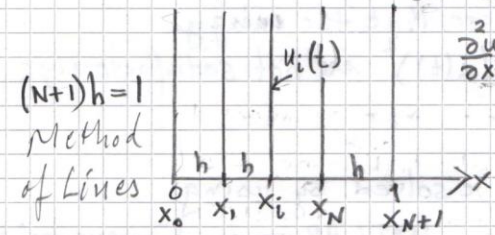
IC: $u(x,0) = u_0(x)$

BC: $u(0,t) = \alpha(t)$, $u(1,t) = \beta(t)$

$\Rightarrow u = u(x,t)$



Approximate $\frac{\partial^2 u}{\partial x^2}$ by central difference ($K = 1$)



$$\frac{\partial^2 u}{\partial x^2}(x_i, t) \approx \frac{u_{i+1}(t) - 2u_i(t) + u_{i-1}(t)}{h^2}$$

$i = 1, 2, \ldots, N$

$u_0(t) = \alpha(t)$

$u_{N+1}(t) = \beta(t)$

$u_i(0) = u_0(x_i)$

ODE system: $\frac{d\bar{u}}{dt} = \frac{1}{h^2} \underbrace{\begin{pmatrix} -2 & 1 & & & 0 \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ 0 & & & 1 & -2 \end{pmatrix}}_{T,\ N \times N} \bar{u} + \frac{1}{h^2} \begin{pmatrix} \alpha(t) \\ 0 \\ \vdots \\ 0 \\ \beta(t) \end{pmatrix}$, $\bar{u}(0) = \begin{pmatrix} u_0(x_1) \\ \vdots \\ u_0(x_N) \end{pmatrix}$ (*)

IVP

The result is not interesting here. More interesting is the number of steps taken. If you run heateqn you see that ode23 needs ~ steps while ode23s needs ~ 50 steps. Why? The problem (*) is stiff. Implicit methods needed like ode23s. ode23 is explicit and has bounded stability region (like explicit Euler).